

Setup: Import project and connect to classicmodels

```
In [1]: import os
import sys
from pathlib import Path

# Make sure Python can find the app folder
project_root = Path.cwd()
if str(project_root) not in sys.path:
    sys.path.append(str(project_root))

# Set MySQL environment variables
# Replace this password if needed on your computer
os.environ["MYSQL_PASSWORD"] = "Db2026!mysql"
os.environ["MYSQL_DATABASE"] = "classicmodels"

print("Notebook setup complete")
```

Notebook setup complete

```
In [3]: import sys

!{sys.executable} -m pip install -r requirements.txt
```

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Collecting fastapi>=0.114.0 (from -r requirements.txt (line 1))
  Using cached fastapi-0.136.1-py3-none-any.whl.metadata (28 kB)
Collecting starlette>=0.38.5 (from -r requirements.txt (line 2))
  Using cached starlette-1.0.0-py3-none-any.whl.metadata (6.3 kB)
Collecting pytest (from -r requirements.txt (line 4))
  Using cached pytest-9.0.3-py3-none-any.whl.metadata (7.6 kB)
Requirement already satisfied: requests in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from -r requirements.txt (line 5)) (2.32.5)
Requirement already satisfied: httpx>=0.27 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from -r requirements.txt (line 6)) (0.28.1)
Requirement already satisfied: pymysql in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from -r requirements.txt (line 7)) (1.1.2)
Requirement already satisfied: cryptography in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from -r requirements.txt (line 8)) (46.0.5)
Requirement already satisfied: notebook in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from -r requirements.txt (line 9)) (7.5.3)
Collecting uvicorn[standard] (from -r requirements.txt (line 3))
  Using cached uvicorn-0.46.0-py3-none-any.whl.metadata (6.7 kB)
Collecting pydantic>=2.9.0 (from fastapi>=0.114.0->-r requirements.txt (line 1))
  Using cached pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
Requirement already satisfied: typing-extensions>=4.8.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from fastapi>=0.114.0->-r requirements.txt (line 1)) (4.15.0)
Collecting typing-inspection>=0.4.2 (from fastapi>=0.114.0->-r requirements.txt (line 1))
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi>=0.114.0->-r requirements.txt (line 1))
  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Requirement already satisfied: anyio<5,>=3.6.2 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from starlette>=0.38.5->-r requirements.txt (line 2)) (4.12.1)
Requirement already satisfied: idna>=2.8 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from anyio<5,>=3.6.2->starlette>=0.38.5->-r requirements.txt (line 2)) (3.11)
Collecting click>=7.0 (from uvicorn[standard]->-r requirements.txt (line 3))
  Using cached click-8.3.3-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: h11>=0.8 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from uvicorn[standard]->-r requirements.txt (line 3)) (0.16.0)
Collecting httptools>=0.6.3 (from uvicorn[standard]->-r requirements.txt (line 3))
  Using cached httptools-0.7.1-cp314-cp314-macosx_11_0_arm64.whl.metadata (3.5 kB)
Collecting python-dotenv>=0.13 (from uvicorn[standard]->-r requirements.txt (line 3))
  Using cached python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)
Requirement already satisfied: pyyaml>=5.1 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from uvicorn[standard]->-r requirements.txt (line 3)) (6.0.3)

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Collecting uvloop>=0.15.1 (from uvicorn[standard]->-r requirements.txt (line
3))
  Using cached uvloop-0.22.1-cp314-cp314-macosx_10_13_universal2.whl.metadata
a (4.9 kB)
Collecting watchfiles>=0.20 (from uvicorn[standard]->-r requirements.txt (li
ne 3))
  Using cached watchfiles-1.1.1-cp314-cp314-macosx_11_0_arm64.whl.metadata
(4.9 kB)
Collecting websockets>=10.4 (from uvicorn[standard]->-r requirements.txt (li
ne 3))
  Using cached websockets-16.0-cp314-cp314-macosx_11_0_arm64.whl.metadata
(6.8 kB)
Collecting iniconfig>=1.0.1 (from pytest->-r requirements.txt (line 4))
  Using cached iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: packaging>=22 in /Library/Frameworks/Python.f
ramework/Versions/3.14/lib/python3.14/site-packages (from pytest->-r requir
ements.txt (line 4)) (26.0)
Collecting pluggy<2,>=1.5 (from pytest->-r requirements.txt (line 4))
  Using cached pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Requirement already satisfied: pygments>=2.7.2 in /Library/Frameworks/Pytho
n.framework/Versions/3.14/lib/python3.14/site-packages (from pytest->-r requ
irements.txt (line 4)) (2.19.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /Library/Framework
s/Python.framework/Versions/3.14/lib/python3.14/site-packages (from request
s->-r requirements.txt (line 5)) (3.4.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in /Library/Frameworks/Pyt
hon.framework/Versions/3.14/lib/python3.14/site-packages (from requests->-r
requirements.txt (line 5)) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in /Library/Frameworks/Pyt
hon.framework/Versions/3.14/lib/python3.14/site-packages (from requests->-r
requirements.txt (line 5)) (2026.1.4)
Requirement already satisfied: httpcore==1.* in /Library/Frameworks/Python.f
ramework/Versions/3.14/lib/python3.14/site-packages (from httpx>=0.27->-r re
quirements.txt (line 6)) (1.0.9)
Requirement already satisfied: cffi>=2.0.0 in /Library/Frameworks/Python.fra
mework/Versions/3.14/lib/python3.14/site-packages (from cryptography->-r req
uirements.txt (line 8)) (2.0.0)
Requirement already satisfied: jupyter-server<3,>=2.4.0 in /Library/Framework
s/Python.framework/Versions/3.14/lib/python3.14/site-packages (from notebook
->-r requirements.txt (line 9)) (2.17.0)
Requirement already satisfied: jupyterlab-server<3,>=2.28.0 in /Library/Fram
eworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from not
ebook->-r requirements.txt (line 9)) (2.28.0)
Requirement already satisfied: jupyterlab<4.6,>=4.5.3 in /Library/Framework
s/Python.framework/Versions/3.14/lib/python3.14/site-packages (from notebook
->-r requirements.txt (line 9)) (4.5.3)
Requirement already satisfied: notebook-shim<0.3,>=0.2 in /Library/Framework
s/Python.framework/Versions/3.14/lib/python3.14/site-packages (from notebook
->-r requirements.txt (line 9)) (0.2.4)
Requirement already satisfied: tornado>=6.2.0 in /Library/Frameworks/Python.
framework/Versions/3.14/lib/python3.14/site-packages (from notebook->-r requ
irements.txt (line 9)) (6.5.4)
Requirement already satisfied: argon2-cffi>=21.1 in /Library/Frameworks/Pyth
on.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server
<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (25.1.0)
Requirement already satisfied: jinja2>=3.0.3 in /Library/Frameworks/Python.f

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ramework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>
=2.4.0->notebook->-r requirements.txt (line 9)) (3.1.6)
Requirement already satisfied: jupyter-client>=7.4.4 in /Library/Frameworks/
Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-se
rver<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (8.8.0)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (5.9.1)
Requirement already satisfied: jupyter-events>=0.11.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.12.0)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.5.4)
Requirement already satisfied: nbconvert>=6.4.4 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (7.17.0)
Requirement already satisfied: nbformat>=5.3.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (5.10.4)
Requirement already satisfied: prometheus-client>=0.9 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.24.1)
Requirement already satisfied: pyzmq>=24 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (27.1.0)
Requirement already satisfied: send2trash>=1.8.2 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (2.1.0)
Requirement already satisfied: terminado>=0.8.3 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.18.1)
Requirement already satisfied: traitlets>=5.6.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (5.14.3)
Requirement already satisfied: websocket-client>=1.7 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (1.9.0)
Requirement already satisfied: async-lru>=1.0.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (2.1.0)
Requirement already satisfied: ipykernel!=6.30.0,>=6.5.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (7.2.0)
Requirement already satisfied: jupyter-lsp>=2.0.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (2.3.0)
Requirement already satisfied: setuptools>=41.1.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (81.0.0)
Requirement already satisfied: babel>=2.10 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyterlab-server<3,>=2.28.0->notebook->-r requirements.txt (line 9)) (2.18.0)
Requirement already satisfied: json5>=0.9.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyterlab-server<3,>=2.28.0->notebook->-r requirements.txt (line 9)) (0.13.0)

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Requirement already satisfied: jsonschema>=4.18.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyterlab-server<3,>=2.28.0->notebook->-r requirements.txt (line 9)) (4.26.0)

Requirement already satisfied: argon2-cffi-bindings in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (25.1.0)

Requirement already satisfied: pycparser in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from cffi>=2.0.0->cryptograph->-r requirements.txt (line 8)) (3.0)

Requirement already satisfied: appnope>=0.1.2 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (0.1.4)

Requirement already satisfied: comm>=0.1.1 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (0.2.3)

Requirement already satisfied: debugpy>=1.6.5 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (1.8.20)

Requirement already satisfied: ipython>=7.23.1 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (9.10.0)

Requirement already satisfied: matplotlib-inline>=0.1 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (0.2.1)

Requirement already satisfied: nest-asyncio>=1.4 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (1.6.0)

Requirement already satisfied: psutil>=5.7 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (7.2.2)

Requirement already satisfied: decorator>=4.3.2 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (5.2.1)

Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (1.1.1)

Requirement already satisfied: jedi>=0.18.1 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (0.19.2)

Requirement already satisfied: pexpect>4.3 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (4.9.0)

Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ip

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python>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->
-r requirements.txt (line 9)) (3.0.52)
Requirement already satisfied: stack_data>=0.6.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (0.6.3)
Requirement already satisfied: wcwidth in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (0.6.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jedi>=0.18.1->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-r requirements.txt (line 9)) (0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jinja2>=3.0.3->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (3.0.3)
Requirement already satisfied: attrs>=22.2.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook->-r requirements.txt (line 9)) (25.4.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook->-r requirements.txt (line 9)) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook->-r requirements.txt (line 9)) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook->-r requirements.txt (line 9)) (0.30.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-client>=7.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (2.9.0.post0)
Requirement already satisfied: platformdirs>=2.5 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (4.5.1)
Requirement already satisfied: python-json-logger>=2.0.4 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (4.0.0)
Requirement already satisfied: rfc3339-validator in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.1.1)
Requirement already satisfied: fqdn in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requi

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requirements.txt (line 9)) (1.5.1)
Requirement already satisfied: isoduration in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (20.11.0)
Requirement already satisfied: jsonpointer>1.13 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (3.0.0)
Requirement already satisfied: rfc3987-syntax>=1.1.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (1.1.0)
Requirement already satisfied: uri-template in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (1.3.0)
Requirement already satisfied: webcolors>=24.6.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (25.10.0)
Requirement already satisfied: beautifulsoup4 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (4.14.3)
Requirement already satisfied: bleach!=5.0.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (6.3.0)
Requirement already satisfied: defusedxml in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.3.0)
Requirement already satisfied: mistune<4,>=2.0.3 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (3.2.0)
Requirement already satisfied: nbclient>=0.5.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.10.4)
Requirement already satisfied: pandocfilters>=1.4.1 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (1.5.1)
Requirement already satisfied: webencodings in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (1.4.0)
Requirement already satisfied: fastjsonschema>=2.15 in /Library/Frameworks/P

```

```

python.framework/Versions/3.14/lib/python3.14/site-packages (from nbformat>=
5.3.0->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (2.
21.2)
Requirement already satisfied: ptyprocess>=0.5 in /Library/Frameworks/Pytho
n.framework/Versions/3.14/lib/python3.14/site-packages (from pexpect>4.3->ip
ython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->
-r requirements.txt (line 9)) (0.7.0)
Collecting annotated-types>=0.6.0 (from pydantic>=2.9.0->fastapi>=0.114.0->-
r requirements.txt (line 1))
  Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.4 (from pydantic>=2.9.0->fastapi>=0.114.0->-r
requirements.txt (line 1))
  Using cached pydantic_core-2.46.4-cp314-cp314-macosx_11_0_arm64.whl.metada
ta (6.6 kB)
Requirement already satisfied: six>=1.5 in /Library/Frameworks/Python.framew
ork/Versions/3.14/lib/python3.14/site-packages (from python-dateutil>=2.8.2-
>jupyter-client>=7.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.
txt (line 9)) (1.17.0)
Requirement already satisfied: lark>=1.2.2 in /Library/Frameworks/Python.fra
mework/Versions/3.14/lib/python3.14/site-packages (from rfc3987-syntax>=1.1.
0->jschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server
<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (1.3.1)
Requirement already satisfied: executing>=1.2.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.14/lib/python3.14/site-packages (from stack_data>=0.
6.0->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->not
ebook->-r requirements.txt (line 9)) (2.2.1)
Requirement already satisfied: asttokens>=2.1.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.14/lib/python3.14/site-packages (from stack_data>=0.
6.0->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->not
ebook->-r requirements.txt (line 9)) (3.0.1)
Requirement already satisfied: pure-eval in /Library/Frameworks/Python.frame
work/Versions/3.14/lib/python3.14/site-packages (from stack_data>=0.6.0->ipy
thon>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.3->notebook->-
r requirements.txt (line 9)) (0.2.3)
Requirement already satisfied: soupsieve>=1.6.1 in /Library/Frameworks/Pytho
n.framework/Versions/3.14/lib/python3.14/site-packages (from beautifulsoup4-
>nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook->-r requirements.txt
(line 9)) (2.8.3)
Requirement already satisfied: arrow>=0.15.0 in /Library/Frameworks/Python.f
ramework/Versions/3.14/lib/python3.14/site-packages (from isoduration->jsons
chema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.
4.0->notebook->-r requirements.txt (line 9)) (1.4.0)
Requirement already satisfied: tzdata in /Library/Frameworks/Python.framewor
k/Versions/3.14/lib/python3.14/site-packages (from arrow>=0.15.0->isoduratio
n->jschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server
<3,>=2.4.0->notebook->-r requirements.txt (line 9)) (2025.3)
Using cached fastapi-0.136.1-py3-none-any.whl (117 kB)
Using cached starlette-1.0.0-py3-none-any.whl (72 kB)
Using cached uvicorn-0.46.0-py3-none-any.whl (70 kB)
Using cached pytest-9.0.3-py3-none-any.whl (375 kB)
Using cached pluggy-1.6.0-py3-none-any.whl (20 kB)
Using cached annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Using cached click-8.3.3-py3-none-any.whl (110 kB)
Using cached httptools-0.7.1-cp314-cp314-macosx_11_0_arm64.whl (108 kB)
Using cached iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Using cached pydantic-2.13.4-py3-none-any.whl (472 kB)

```



```
Using cached pydantic_core-2.46.4-cp314-cp314-macosx_11_0_arm64.whl (2.0 MB)
Using cached annotated_types-0.7.0-py3-none-any.whl (13 kB)
Using cached python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Using cached typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Using cached uvloop-0.22.1-cp314-cp314-macosx_10_13_universal2.whl (1.4 MB)
Using cached watchfiles-1.1.1-cp314-cp314-macosx_11_0_arm64.whl (390 kB)
Using cached websockets-16.0-cp314-cp314-macosx_11_0_arm64.whl (175 kB)
Installing collected packages: websockets, uvloop, typing-inspection, python-dotenv, pydantic-core, pluggy, iniconfig, httpptools, click, annotated-types, annotated-doc, watchfiles, uvicorn, starlette, pytest, pydantic, fastapi
17/17 [fastapi]- 15/17 [pydantic]
c]test]tel]
Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0 click-8.3.3 fastapi-0.136.1 httpptools-0.7.1 iniconfig-2.3.0 pluggy-1.6.0 pydantic-2.13.4 pydantic-core-2.46.4 pytest-9.0.3 python-dotenv-1.2.2 starlette-1.0.0 typing-inspection-0.4.2 uvicorn-0.46.0 uvloop-0.22.1 watchfiles-1.1.1 websockets-16.0
```

[notice] A new release of pip is available: 25.2 -> 26.1.1

[notice] To update, run: /Library/Frameworks/Python.framework/Versions/3.14/bin/python3 -m pip install --upgrade pip

Test 1: Read one customer from MySQL

```
In [16]: from app.resources.CustomerResource import CustomerResource
```

```
customer_resource = CustomerResource()

customer = customer_resource.get_by_id("112")
customer
```

```
Out[16]: Customer(customerNumber=112, customerName='Signal Gift Stores', contactLastName='King', contactFirstName='Jean', phone='7025551838', addressLine1='848 9 Strong St.', addressLine2=None, city='Las Vegas', state='NV', postalCode='83030', country='USA', salesRepEmployeeNumber=1166, creditLimit=Decimal('71800.00'))
```

Test 2: Read one order from MySQL

```
In [17]: from app.resources.OrderResource import OrderResource
```

```
order_resource = OrderResource()

order = order_resource.get_by_id("10101")
order
```

```
Out[17]: Order(orderNumber=10101, orderDate=datetime.date(2003, 1, 9), requiredDate=datetime.date(2003, 1, 18), shippedDate=datetime.date(2003, 1, 11), status='Shipped', comments='Check on availability.', customerNumber=128)
```

Test 3: Read all order details for one order

```
In [18]: from app.resources.OrderDetailsResource import OrderDetailsResource

order_details_resource = OrderDetailsResource()

order_details = order_details_resource.get_by_id("10101")
order_details
```

```
Out[18]: OrderDetailsCollection(items=[OrderDetail(orderNumber=10101, productCode='S18_2325', quantityOrdered=25, priceEach=Decimal('108.06'), orderLineNumber=4), OrderDetail(orderNumber=10101, productCode='S18_2795', quantityOrdered=26, priceEach=Decimal('167.06'), orderLineNumber=1), OrderDetail(orderNumber=10101, productCode='S24_1937', quantityOrdered=45, priceEach=Decimal('32.53'), orderLineNumber=3), OrderDetail(orderNumber=10101, productCode='S24_2022', quantityOrdered=46, priceEach=Decimal('44.35'), orderLineNumber=2)])
```

Test 4: Read one exact order detail using the composite key

```
In [19]: exact_order_detail = order_details_resource.get_by_order_and_product(
        "10101",
        "S18_2325"
    )

exact_order_detail
```

```
Out[19]: OrderDetail(orderNumber=10101, productCode='S18_2325', quantityOrdered=25, priceEach=Decimal('108.06'), orderLineNumber=4)
```

Test 5: Update one exact order detail row using the composite key

```
In [20]: from app.resources.OrderDetailsResource import OrderDetail

updated_detail = OrderDetail(
    orderNumber=10101,
    productCode="S18_2325",
    quantityOrdered=26,
    priceEach="108.06",
    orderLineNumber=4,
)

updated_count = order_details_resource.put_by_order_and_product(
    "10101",
    "S18_2325",
    updated_detail,
)

print("Rows updated:", updated_count)

order_details_resource.get_by_order_and_product("10101", "S18_2325")
```

Rows updated: 1

```
Out[20]: OrderDetail(orderNumber=10101, productCode='S18_2325', quantityOrdered=26,  
priceEach=Decimal('108.06'), orderLineNumber=4)
```

Test 6: Restore the original order detail value

```
In [21]: restored_detail = OrderDetail(  
    orderNumber=10101,  
    productCode="S18_2325",  
    quantityOrdered=25,  
    priceEach="108.06",  
    orderLineNumber=4,  
)  
  
restored_count = order_details_resource.put_by_order_and_product(  
    "10101",  
    "S18_2325",  
    restored_detail,  
)  
  
print("Rows restored:", restored_count)  
  
order_details_resource.get_by_order_and_product("10101", "S18_2325")
```

Rows restored: 1

```
Out[21]: OrderDetail(orderNumber=10101, productCode='S18_2325', quantityOrdered=25,  
priceEach=Decimal('108.06'), orderLineNumber=4)
```

Final Summary of what I did!

```
In [22]: print("The task 6 demo is complete!")  
print("I tested CustomerResource get_by_id with customerNumber 112.")  
print("I tested OrderResource get_by_id with orderNumber 10101.")  
print("I tested OrderDetailsResource get_by_id with orderNumber 10101.")  
print("I tested composite-key get for orderdetails using orderNumber 10101 and productCode S18_2325.")  
print("I tested composite-key update and restored the original value.")
```

The task 6 demo is complete!

I tested CustomerResource get_by_id with customerNumber 112.

I tested OrderResource get_by_id with orderNumber 10101.

I tested OrderDetailsResource get_by_id with orderNumber 10101.

I tested composite-key get for orderdetails using orderNumber 10101 and productCode S18_2325.

I tested composite-key update and restored the original value.